AI 2002: Artificial Intelligence

Quantifying Uncertainty

Probabilistic Reasoning

Spring 2026

[These slides (mostly) are based on those created by Dan Klein and Pieter Abbeel (ai.berkeley.edu).]

[Some slides are by Mr. Sandesh Kumar, and Ms. Anaum Hamid, FAST-NUCES, Karahi, and by Nitin Agarwal, Software Consultant]

Probability

- Random Variables
- Joint and Marginal Distributions
- Conditional Distribution
- Product Rule, Chain Rule, Bayes' Rule
- Inference
- Independence

Uncertainty

- Probabilities are a way to summarize uncertainty
  - With the facts I can observe ...
  - There's 80% of chance of having a cavity

- General situation:

  - **Observed variables (evidence)**: Agent knows certain things about the state of the world (e.g., sensor readings or symptoms)

  - **Unobserved variables**: Agent needs to reason about other aspects (e.g. where an object is or what disease is present)

  - **Model**: Agent knows something about how the known variables relate to the unknown variables

- Probabilistic reasoning gives us a framework for managing our beliefs and knowledge

Decision Theory

Likelihood and Importance

- Rational decisions involve:
  - Likelihood of achieving a state/goal (probability)
  - Relative importance of the states/goals (Utility)

- Decision Theory = probability theory + utility theory

- Maximum Expected Utility
  - An agent is rational iff it chooses the action that yields the highest expected utility, averaged over all the possible outcomes of the action.

Random Variables

- A random variable is some aspect of the world about which we (may) have uncertainty
  - R = Is it raining?
  - T = Is it hot or cold?
  - D = How long will it take to drive to work?
  - L = Where is the ghost?
- We denote random variables with capital letters
- Like variables in a CSP, random variables have domains
  - T in {hot, cold}
  - L in possible locations, maybe {(0,0), (0,1), …}

Probability Distributions

- Associate a probability with each value

  - Temperature:

$$P(T)$$

T	P
hot	0.5
cold	0.5

  - Weather:

$$P(W)$$

W	P
sun	0.6
rain	0.1
fog	0.3
meteor	0.0

Probability Distributions

- Unobserved random variables have distributions

$P(T)$

T	P
hot	0.5
cold	0.5

$P(W)$

W	P
sun	0.6
rain	0.1
fog	0.3
meteor	0.0

- A distribution is a TABLE of probabilities of values

Shorthand notation:

$$P(hot) = P(T = hot),$$
$$P(cold) = P(T = cold),$$
$$P(rain) = P(W = rain),$$
$$\ldots$$

OK *if* all domain entries are unique

- A probability (lower case value) is a single number

$$P(W = rain) = 0.1$$

- Must have: $\forall x \;\; P(X = x) \geq 0$ and $\sum_x P(X = x) = 1$

Joint Distributions

- A *joint distribution* over a set of random variables: $X_1, X_2, \ldots X_n$ specifies a real number for each assignment (or *outcome*):

$$P(X_1 = x_1, X_2 = x_2, \ldots X_n = x_n)$$

$$P(x_1, x_2, \ldots x_n)$$

  - Must obey: $$P(x_1, x_2, \ldots x_n) \geq 0$$

$$\sum_{(x_1, x_2, \ldots x_n)} P(x_1, x_2, \ldots x_n) = 1$$

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

- Size of distribution if n variables with domain sizes d?
  - For all but the smallest distributions, impractical to write out!

Probabilistic Models

- A probabilistic model is a **joint distribution** over a set of random variables

- Probabilistic models:
  - (Random) variables with domains
  - Assignments are called *outcomes*
  - *Normalized:* sum to 1.0

Distribution over T,W

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Constraint over T,W

T	W	P
hot	sun	T
hot	rain	F
cold	sun	F
cold	rain	T

Events

- An *event* is a set E of outcomes

$$P(E) = \sum_{(x_1 \dots x_n) \in E} P(x_1 \dots x_n)$$

- From a joint distribution, we can calculate the probability of any event
  - Probability that it's hot AND sunny?
  - Probability that it's hot?
  - Probability that it's hot OR sunny?
- Typically, the events we care about are *partial assignments*, like P(T=hot)

$$P(T, W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Events

- P(+x, +y) ?

- P(+x) ?

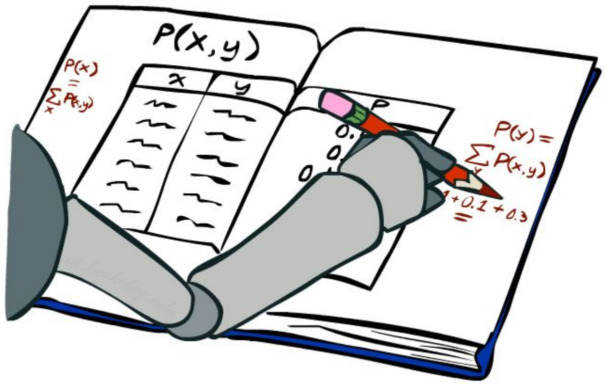
- P(-y OR +x) ?

$$P(X, Y)$$

X	Y	P
+x	+y	0.2
+x	-y	0.3
-x	+y	0.4
-x	-y	0.1

Marginal Distributions

- Marginal distributions are sub-tables which eliminate variables
- Marginalization (summing out): Combine collapsed rows by adding



$$P(T,W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$P(t) = \sum_s P(t,s)$$

$$P(s) = \sum_t P(t,s)$$

$$P(T)$$

T	P
hot	0.5
cold	0.5

$$P(W)$$

W	P
sun	0.6
rain	0.4

$$P(X_1 = x_1) = \sum_{x_2} P(X_1 = x_1, X_2 = x_2)$$

Quiz: Marginal Distributions

$$P(X,Y)$$

X	Y	P
+x	+y	0.2
+x	-y	0.3
-x	+y	0.4
-x	-y	0.1

$$P(x) = \sum_y P(x,y)$$

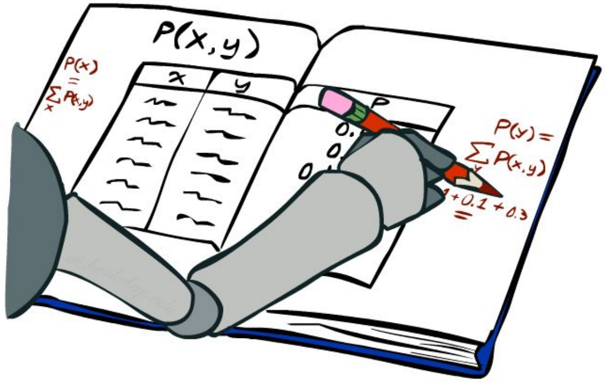
$$P(y) = \sum_x P(x,y)$$

$$P(X)$$

X	P
+x	
-x	

$$P(Y)$$

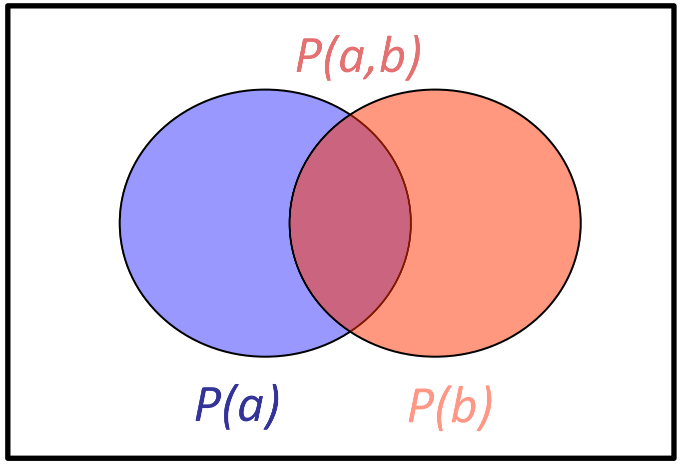
Y	P
+y	
-y	



Conditional Probabilities

- A simple relation between joint and conditional probabilities
  - In fact, this is taken as the *definition* of a conditional probability

$$P(a|b) = \frac{P(a,b)}{P(b)}$$



$$P(T,W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$P(W=s|T=c) = \frac{P(W=s,T=c)}{P(T=c)} = \frac{0.2}{0.5} = 0.4$$

$$= P(W=s,T=c) + P(W=r,T=c)$$
$$= 0.2 + 0.3 \quad = 0.5$$

Conditional Probabilities

- 5% of people have the disease
  - P(D) = 0.05
  - $P(\neg D)$ = 0.95

Disease (D)	Test Result (T)	Probability
D	T	0.045
D	¬T	0.005
¬D	T	0.095
¬D	¬T	0.855

- The test detects the disease 90% of the time (sensitivity)
  - $P(T \mid D)$ = 0.9
  - $P(T \mid \neg D)$ = 0.1
  - $P(T \mid D)$ = P (T, D) / P(D) = 0.045 / 0.05 = 0.9
  - P(T) = 0.14
  - P(¬T) = 0.86

$$P(D \mid T) = \frac{0.045}{0.14} \approx 0.32$$

Quiz: Conditional Probabilities

$P(X, Y)$

X	Y	P
+x	+y	0.2
+x	-y	0.3
-x	+y	0.4
-x	-y	0.1

- P(+x | +y) ?
- P(-x | +y) ?
- P(-y | +x) ?

Conditional Distributions

- Conditional distributions are probability distributions over some variables given fixed values of others

Conditional Distributions

$P(W|T)$

$P(W|T = hot)$

W	P
sun	0.8
rain	0.2

$P(W|T = cold)$

W	P
sun	0.4
rain	0.6

Joint Distribution

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Conditional Distributions

$$P(T, W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$\begin{aligned} P(W=s|T=c) &= \frac{P(W=s,T=c)}{P(T=c)} \\ &= \frac{P(W=s,T=c)}{P(W=s,T=c)+P(W=r,T=c)} \\ &= \frac{0.2}{0.2+0.3} = 0.4 \end{aligned}$$

$$\longrightarrow$$

$$\begin{aligned} P(W=r|T=c) &= \frac{P(W=r,T=c)}{P(T=c)} \\ &= \frac{P(W=r,T=c)}{P(W=s,T=c)+P(W=r,T=c)} \\ &= \frac{0.3}{0.2+0.3} = 0.6 \end{aligned}$$

$$P(W|T=c)$$

W	P
sun	0.4
rain	0.6

Normalization Trick

$$
\begin{aligned}
P(W=s|T=c) &= \frac{P(W=s,T=c)}{P(T=c)} \\
&= \frac{P(W=s,T=c)}{P(W=s,T=c)+P(W=r,T=c)} \\
&= \frac{0.2}{0.2+0.3} = 0.4
\end{aligned}
$$

$P(T,W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

SELECT the joint probabilities matching the evidence

$P(c,W)$

T	W	P
cold	sun	0.2
cold	rain	0.3

NORMALIZE the selection (make it sum to one)

$P(W|T=c)$

W	P
sun	0.4
rain	0.6

$$
\begin{aligned}
P(W=r|T=c) &= \frac{P(W=r,T=c)}{P(T=c)} \\
&= \frac{P(W=r,T=c)}{P(W=s,T=c)+P(W=r,T=c)} \\
&= \frac{0.3}{0.2+0.3} = 0.6
\end{aligned}
$$

Normalization Trick

$$P(T, W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

SELECT the joint probabilities matching the evidence →

$$P(c, W)$$

T	W	P
cold	sun	0.2
cold	rain	0.3

NORMALIZE the selection (make it sum to one) →

$$P(W|T = c)$$

W	P
sun	0.4
rain	0.6

- Why does this work? Sum of selection is P(evidence)! (P(T=c), here)

$$P(x_1|x_2) = \frac{P(x_1, x_2)}{P(x_2)} = \frac{P(x_1, x_2)}{\sum_{x_1} P(x_1, x_2)}$$

To Normalize

- (Dictionary) To bring or restore to a normal condition

All entries sum to ONE

- Procedure:
  - Step 1: Compute Z = sum over all entries
  - Step 2: Divide every entry by Z

- Example 1

W	P
sun	0.2
rain	0.3

Normalize

Z = 0.5

W	P
sun	0.4
rain	0.6

- Example 2

T	W	P
hot	sun	20
hot	rain	5
cold	sun	10
cold	rain	15

Normalize

Z = 50

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Probabilistic Inference

- Compute a desired probability from other known probabilities (e.g. conditional from joint)

- We generally compute conditional probabilities
  - P(on time | no reported accidents) = 0.90
  - These represent the agent's *beliefs* given the evidence

- Probabilities change with new evidence:
  - P(on time | no accidents, 5 a.m.) = 0.95
  - P(on time | no accidents, 5 a.m., raining) = 0.80
  - Observing new evidence causes *beliefs to be updated*

Inference by Enumeration

- General case:
  - Evidence variables: $E_1 \dots E_k = e_1 \dots e_k$
  - Query* variable: Q
  - Hidden variables: $H_1 \dots H_r$

$X_1, X_2, \dots X_n$

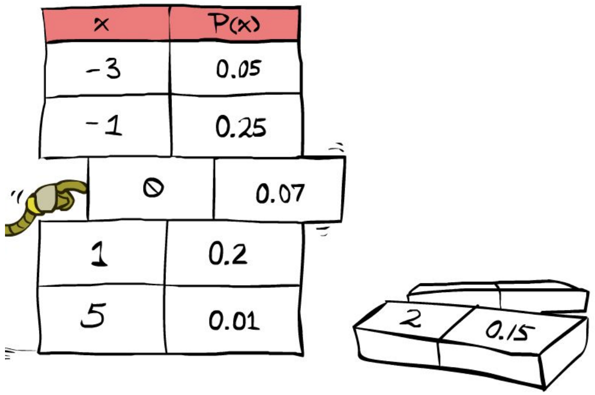
All variables

- We want:

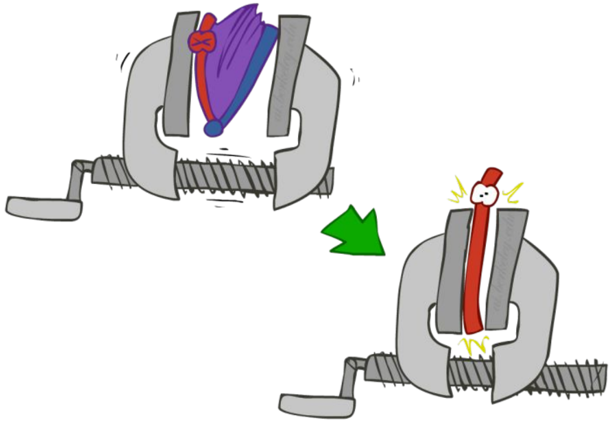
** Works fine with multiple query variables, too*

$$P(Q|e_1 \dots e_k)$$

- Step 1: Select the entries consistent with the evidence



- Step 2: Sum out H to get joint of Query and evidence



$$P(Q, e_1 \dots e_k) = \sum_{h_1 \dots h_r} \underbrace{P(Q, h_1 \dots h_r, e_1 \dots e_k)}_{X_1, X_2, \dots X_n}$$

- Step 3: Normalize

$$\times \frac{1}{Z}$$

$$Z = \sum_q P(Q, e_1 \cdots e_k)$$

$$P(Q|e_1 \cdots e_k) = \frac{1}{Z} P(Q, e_1 \cdots e_k)$$

Inference by Enumeration

- P(W)?

- P(W | winter)?

- P(W | winter, hot)?

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Inference by Enumeration

- P(W)?

- P(W | winter)?

W	
S	0.25
R	0.25

- P(W | winter, hot)?

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

1

Inference by Enumeration

- Obvious problems:
  - Worst-case time complexity $O(d^n)$
  - Space complexity $O(d^n)$ to store the joint distribution

The Product Rule

$$P(y)P(x|y) = P(x,y)$$

- Example:

$P(W)$

R	P
sun	0.8
rain	0.2

$P(D|W)$

D	W	P
wet	sun	0.1
dry	sun	0.9
wet	rain	0.7
dry	rain	0.3

$P(D,W)$

D	W	P
wet	sun	
dry	sun	
wet	rain	
dry	rain	

The Chain Rule

- More generally, can always write any joint distribution as an incremental product of conditional distributions

$$P(x_1, x_2, x_3) = P(x_1)P(x_2|x_1)P(x_3|x_1, x_2)$$

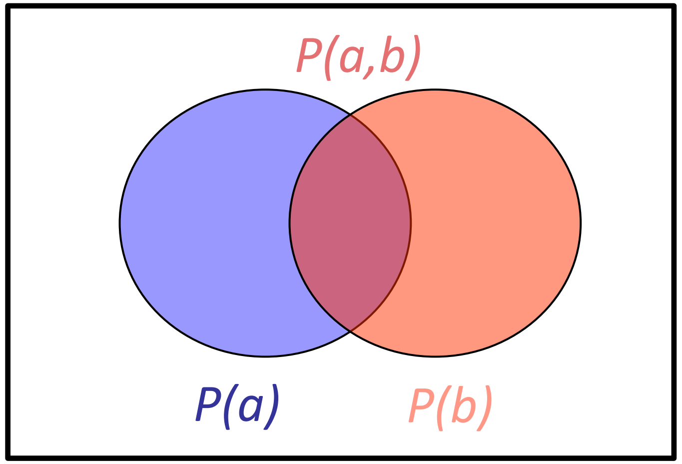
$$P(x_1, x_2, \ldots x_n) = \prod_i P(x_i|x_1 \ldots x_{i-1})$$

- Why is this always true?

Review: Conditional Probabilities

- A simple relation between joint and conditional probabilities
  - In fact, this is taken as the *definition* of a conditional probability

$$P(a|b) = \frac{P(a,b)}{P(b)}$$



$$P(T,W)$$

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$$P(W=s|T=c) = \frac{P(W=s,T=c)}{P(T=c)} = \frac{0.2}{0.5} = 0.4$$

$$= P(W=s,T=c) + P(W=r,T=c)$$
$$= 0.2 + 0.3 = 0.5$$

Bayes' Rule

- Two ways to factor a joint distribution over two variables:

$$P(x, y) = P(x|y)P(y) = P(y|x)P(x)$$

- Dividing, we get:

$$P(x|y) = \frac{P(y|x)}{P(y)}P(x)$$

- Why is this at all helpful?
  - Lets us build one conditional from its reverse
  - Often one conditional is tricky but the other one is simple
  - Foundation of many systems we'll see later (e.g. ASR, MT)

- In the running for most important AI equation!

Bayes' Rule

Disease (D)	Test Result (T)	Probability
D	T	0.045
D	¬T	0.005
¬D	T	0.095
¬D	¬T	0.855

- 5% of people have the disease
  - P(D) = 0.05
  - $P(\neg D)$ = 0.95
- The test detects the disease 90% of the time (sensitivity)
  - $P(T|D)$ = 0.9
  - $P(T|\neg D)$ = 0.1
  - P(T) = 0.14
  - P(¬T) = 0.86

- $P(T|D)$ = P (T, D) / P(D) = 0.045 / 0.05 = 0.9
- P(D|T) = P (T, D) / P(T) = 0.045 / 0.14 = 0.32
- $P(D|T)$ = P (T|D) * P(D) / P(T) = 0.9 * 0.05 / 0.14 = 0.32

Inference with Bayes' Rule

- Example: Diagnostic probability from causal probability:

$$P(\text{cause}|\text{effect}) = \frac{P(\text{effect}|\text{cause})P(\text{cause})}{P(\text{effect})}$$

- Example:
  - M: meningitis, S: stiff neck

$$\left.\begin{aligned} P(+m) &= 0.0001 \\ P(+s|+m) &= 0.8 \\ P(+s|-m) &= 0.01 \end{aligned}\right\} \text{Example givens}$$

$$P(+m|+s) = \frac{P(+s|+m)P(+m)}{P(+s)} = \frac{P(+s|+m)P(+m)}{P(+s|+m)P(+m) + P(+s|-m)P(-m)} = \frac{0.8 \times 0.0001}{0.8 \times 0.0001 + 0.01 \times 0.999}$$

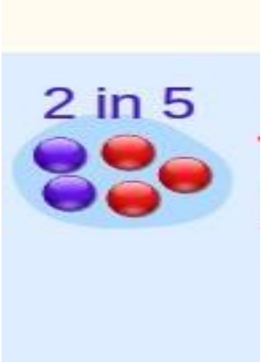
  - Note: posterior probability of meningitis still very small
  - Note: you should still get stiff necks checked out! Why?

Baye’s Rule / Theorem

Revision

Example: Coins in a Bowl (Conditional Probability)

- 2 **blue** and 3 **red** coins are in a bowl.
- What are the chances of getting a **blue** coin?
- The chance is 2 in 5 (i.e. 2/5)



- But after taking one out of these chances, situation may change!
  - if we got a **red** coin before, then the chance of a **blue** coin next is:
  - 2 in 4 (i.e. 2/4)
  - if we got a **blue** coin before, then the chance of a **blue** coin next is:
  - 1 in 4 (i.e. 1/4)

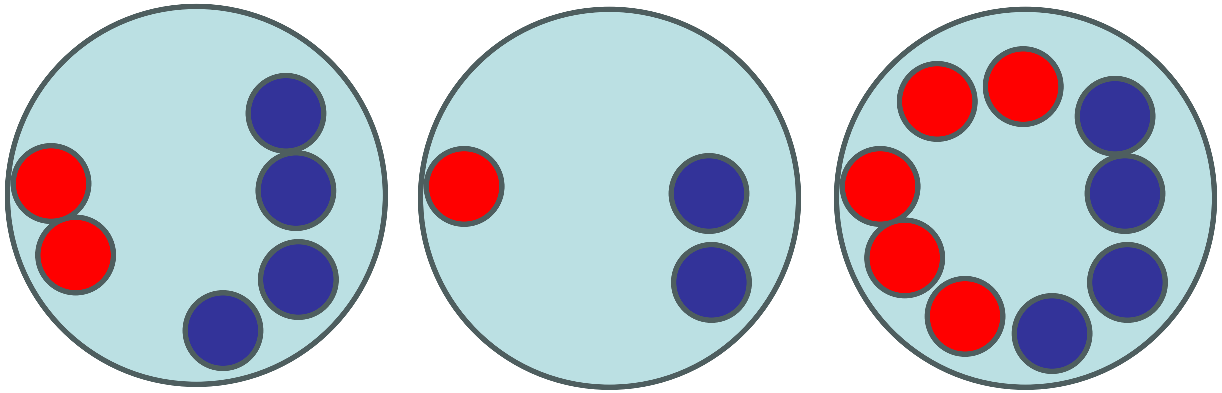
The Formula for Bayes' theorem

$$P(H|E) = \frac{P(E|H) * P(H)}{P(E)}$$

- P(H)
  - the probability of hypothesis H being true. This is known as prior probability.
- P(E)
  - the probability of the evidence(regardless of the hypothesis).
- P(E|H)
  - the probability of the evidence given that hypothesis is true.
- P(H|E)
  - the probability of the hypothesis given that the evidence is there.

Example (Baye's Theorem / Rule)

- Let a red coin was drawn
  - What is probability that
  - it came from bowl 1
  - i.e. **P(B1|R) =?**



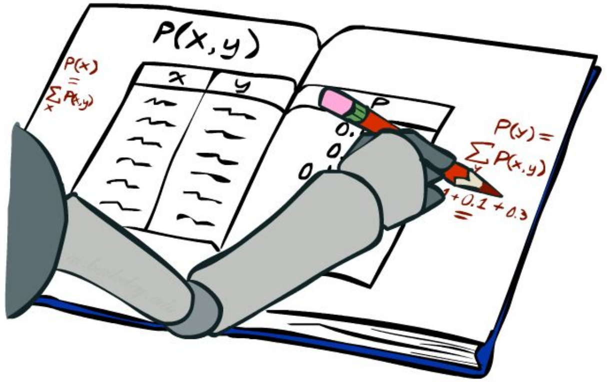
- According to Bayes' theorem

$$P(B1|R) = P(R|B1) * P(B1) / P(R)$$

- We need to know:
  - P(R): Probability to select a red coin
  - P(B1): Probability to select the bowl 1 (B1); **already given: = 1/3**
  - P(R|B1): Probability to select a red coin from B1 **= 2/6**

Review: Marginal Distributions

- Marginal distributions are sub-tables which eliminate variables
- Marginalization (summing out): Combine collapsed rows by adding



$P(T,W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$\mathbf{P(T)} = \sum_W P(T,W)$

$\mathbf{P(W)} = \sum_T P(T,W)$

$P(T)$

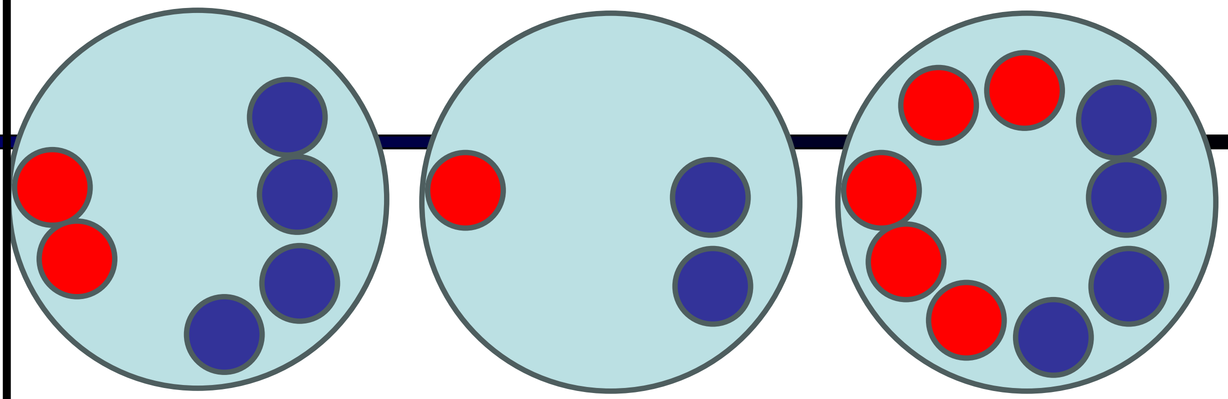
T	P
hot	0.5
cold	0.5

$P(W)$

W	P
sun	0.6
rain	0.4

$$P(X_1 = x_1) = \sum_{x_2} P(X_1 = x_1, X_2 = x_2)$$

Example (Baye's Theorem)



- **P(R) = $\sum_B P(B,R)$**

  = P(B1,R) + (B2,R) + P(B3,R)

  - Where:
  - Probability to select bowl 1 and red coin: P(B1,R) = P (B1) * P(R|B1)
  - Probability to select bowl 2 and red coin: P(B2,R) = P (B2) * P(R|B2)
  - Probability to select bowl 3 and red coin: P(B3,R) = P (B3) * P(R|B3)
- **P(R) = P(B1,R) + (B2,R) + P(B3,R)**
- P(R) = P (B1) * P(R|B1) + P (B2) * P(R|B2) + P(B3) * P(R|B3)
- P(R) = 1/3 * 2/6 + 1/6 * 1/3 + 1/2 * 5/9 = 4/9

. . . .

- **P(B1|R) = P(R|B1) * P(B1) / P(R)**
- P(B1|R) = 2/6 * 1/3 / 4/9 = 2/8 = 0.25
- So if a red coin was drawn then it will be 25% chance that it was drawn from B1

Naive Bayes

Naive Bayes Classifier

- Bayes Theorem $$P(H|E) = \frac{P(E|H) * P(H)}{P(E)}$$
- Naive Bayes is a kind of **classifier** which uses the Bayes Theorem.
- It predicts membership probabilities for each class:
  - such as the probability that given record or data point belongs to a particular class.
  - The class with the highest probability is considered as the most likely class.
  - This is also known as **Maximum A Posteriori (MAP)**.
- MAP(H) = max (P(H|E))

  = max (P(E|H) * P(H) / P(E))

  = max (P(E|H) * P(H))

P(E)= Evidence probability, used to normalize result.
It remains same, removing it, won't affect.

P(H| E1, E2, ... En) = P(E1| H)* P(E2|H) ……*P(En|H) * P(H) / P(E1,E2, ... En)
P(H|Multiple Evidences) = P(E1| H)* P(E2|H) ……*P(En|H) * P(H) / P(Multiple Evidence

Example: Naive Bayes Classifier

- Let's consider a training dataset with 1500 records and 3 classes.
- We presume that there are no missing values in our data.
- We have 3 classes associated with Animal Types:
  - Parrot
  - Dog
  - Fish
- The Predictor features set consists of 4 features as
  - Swim
  - Wings
  - Green Color
  - Dangerous Teeth.
- All features are categorical variables with either of 2 values: T(True) / F(False)

Example: Naive Bayes Classifier

- The table shows a frequency table of our data. In our training data:

Swim	Wings	Green Color	Dangerous Teeth	Animal Type
50	500/500	400/500	0	Parrot
450/500	0	0	500/500	Dog
500/500	0	100/500	50/500	Fish

  - Parrots have 50 (10%) value for Swim, i.e., 10% parrot can swim according to our data, 500 out of 500(100%) parrots have wings, 400 out of 500(80%) parrots are Green and 0(0%) parrots have Dangerous Teeth.
  - Classes with Animal type Dogs shows that 450 out of 500(90%) can swim, 0(0%) dogs have wings, 0(0%) dogs are of Green color and 500 out of 500(100%) dogs have Dangerous Teeth.
  - Classes with Animal type Fishes shows that 500 out of 500(100%) can swim, 0(0%) fishes have wings, 100(20%) fishes are of Green color and 50 out of 500(10%) Fishes have Dangerous Teeth.

Example: Naive Bayes Classifier

- We have taken 2 records that have values in their feature set.
- We have to predict animal type using the feature values. We have to predict whether the animal is a Dog, a Parrot or a Fish

	Swim	Wings	Green	Teeth
1.	True	False	True	False
2.	True	False	True	True

- P(H|Multiple Evidences) = P(E1| H)* P(E2|H) ……*P(En|H) * P(H) / P(Multiple Evidences)
- Let's consider the first record. The Evidence here is Swim & Green.
- The Hypothesis can be an animal type to be Dog, Parrot, Fish.

Example: Naive Bayes Classifier

- For 1st Record
- For Hypothesis testing for the animal to be a Dog:
  - P(Dog | Swim, Green) = P(Swim|Dog) * P(Green|Dog) * P(Dog) / P(Swim, Green)

    = 0.9 * 0 * 0.333 / P(Swim, Green) = 0
- For Hypothesis testing for the animal to be a Parrot:
  - P(Parrot| Swim, Green) = P(Swim|Parrot) * P(Green|Parrot) * P(Parrot) / P(Swim, Green)

    = 0.1 * 0.80 * 0.333 / P(Swim, Green)

    = 0.0264/ P(Swim, Green)
- For Hypothesis testing for the animal to be a Fish:
  - P(Fish| Swim, Green) = P(Swim|Fish) * P(Green|Fish) * P(Fish) / P(Swim, Green)

    = 1 * 0.2 * 0.333 / P(Swim, Green)

    = 0.0666 / P(Swim, Green)
- Using Naive Bayes, we can predict that the class of this record is Fish.

The denominator of all the above calculations is same i.e. P(Swim, Green). The value of P(Fish| Swim, Green) is greater that P(Parrot| Swim, Green).

Let's consider the second record.

The Evidence here is Swim, Green & Teeth. The Hypothesis can be an animal type to be Dog, Parrot, Fish.

For Hypothesis testing for the animal to be a Dog:

P(Dog | Swim, Green, Teeth) = P(Swim|Dog) * P(Green|Dog) * P(Teeth|Dog) * P(Dog) / P(Swim, Green, Teeth)

= 0.9 * 0 * 1 * 0.333 / P(Swim, Green, Teeth) = 0

For Hypothesis testing for the animal to be a Parrot:

P(Parrot| Swim, Green, Teeth) = P(Swim|Parrot) * P(Green|Parrot)* P(Teeth| Parrot) * P(Parrot) / P(Swim, Green, Teeth)

= 0.1 * 0.80 * 0 *0.333 / P(Swim, Green, Teeth) = 0

For Hypothesis testing for the animal to be a Fish:

P(Fish|Swim, Green, Teeth) = P(Swim|Fish) * P(Green|Fish) * P(Teeth|Fish) *P(Fish) / P(Swim, Green, Teeth)

= 1 * 0.2 * 0.1 * 0.333 / P(Swim, Green, Teeth)

= 0.00666 / P(Swim, Green, Teeth)

The value of P(Fish| Swim, Green, Teeth) is the only positive value greater than 0. Using Naive Bayes, we can predict that the class of this record is **Fish.**

Types of Naive Bayes Algorithm

- Gaussian
  - It is used in classification and it assumes that features follow a normal distribution.
- Bernoulli
  - The binomial model is useful if your feature vectors are binary (i.e. 0s and 1s)
  - One application would be text classification with ‘bag of words’ model where the 1s & 0s are “word occurs in the document” and “word does not occur in the document” respectively.
- MultiNomial
  - preferred to use on data that is multinomially distributed.
  - For example, let’s say, we have a text classification problem. Here we can consider bernoulli trials which is one step further and instead of “word occurring in the document”, we have “count how often word occurs in the document”, you can think of it as “number of times outcome number x_i is observed over the n trials”.

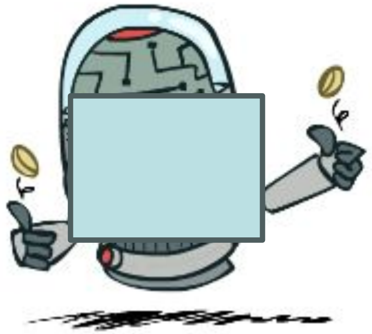
Conditional Independence

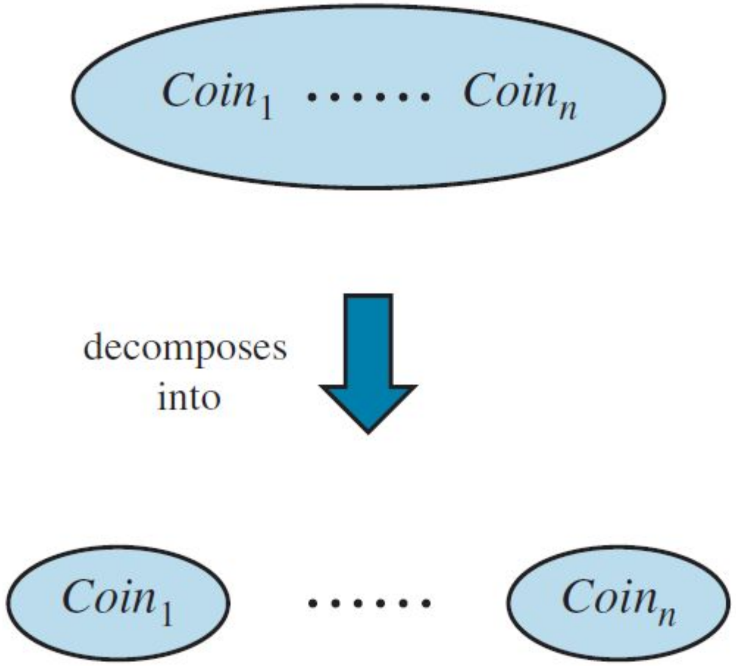
Independence

- Two variables are *independent* if:

$$\forall x, y : P(x, y) = P(x)P(y)$$

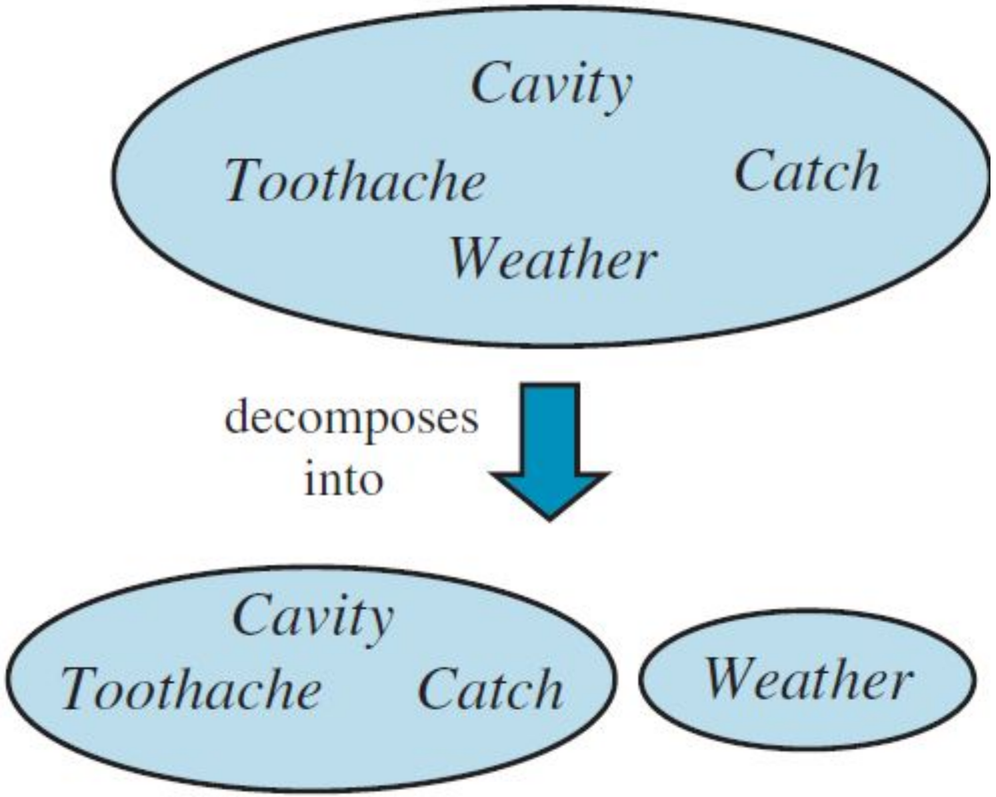
  - This says that their joint distribution *factors* into a product two simpler distributions
  - Another form: $\forall x, y : P(x|y) = P(x)$
  - We write: $X \perp\!\!\!\perp Y$
- Independence is a simplifying *modeling assumption*
- *Independence assumptions are essential for efficient probabilistic reasoning*
  - For n independent biased coins, $O(2^n)$ entries $\rightarrow O(n)$





Independence

- What could we assume for {Weather, Catch, Cavity, Toothache}?



- P(T, X, C, W) = P(T, X, C) * P(W)

Conditional Independence

- P(Toothache, Cavity, Catch)

- If I have a cavity, the probability that the X-ray catches in it doesn't depend on whether I have a toothache:
  - P(+catch | +toothache, +cavity) = P(+catch | +cavity)

- The same independence holds if I don’t have a cavity:
  - P(+catch | +toothache, -cavity) = P(+catch| -cavity)

- Catch is *conditionally independent* of Toothache given Cavity:
  - P(Catch | Toothache, Cavity) = P(Catch | Cavity)

- Equivalent statements:
  - P(Toothache | Catch , Cavity) = P(Toothache | Cavity)
  - P(Toothache, Catch | Cavity) = P(Toothache | Cavity) P(Catch | Cavity)
  - One can be derived from the other easily

Conditional Independence and the Chain Rule

- Chain rule: $$P(X_1, X_2, \ldots X_n) = P(X_1)P(X_2|X_1)P(X_3|X_1, X_2)\ldots$$

- Trivial decomposition:

$$P(\text{Traffic}, \text{Rain}, \text{Umbrella}) = P(\text{Rain})P(\text{Traffic}|\text{Rain})P(\text{Umbrella}|\text{Rain}, \text{Traffic})$$

- With assumption of conditional independence:

$$P(\text{Traffic}, \text{Rain}, \text{Umbrella}) = P(\text{Rain})P(\text{Traffic}|\text{Rain})P(\text{Umbrella}|\text{Rain})$$

- Bayes'nets / graphical models help us express conditional independence assumptions

Conditional Independence

- Unconditional (absolute) independence very rare (why?)
- *Conditional independence* is most basic and robust form of knowledge about uncertain environments.
- **X is conditionally independent of Y given Z**

  if and only if: $X \perp\!\!\!\perp Y | Z$

$$\forall x, y, z : P(x, y|z) = P(x|z)P(y|z)$$

or, equivalently, if and only if

$$\forall x, y, z : P(x|z, y) = P(x|z)$$

Bayes' Nets

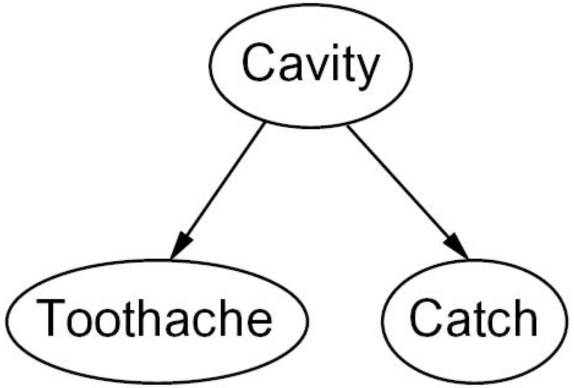
Bayes' Nets: Big Picture

- Two problems with using full joint distribution tables as our probabilistic models:
  - Unless there are only a few variables, the joint is WAY too big to represent explicitly
  - Hard to learn (estimate) anything empirically about more than a few variables at a time

- Bayes' nets: a technique for describing complex joint distributions (models) using simple, local distributions (conditional probabilities)
  - **subset of the general class of graphical models**
- Use local causality/conditional independence:
  - the world is composed of many variables,
  - each interacting locally with a few others
- Outline
  - Representation
  - Exact inference
  - Approximate inference

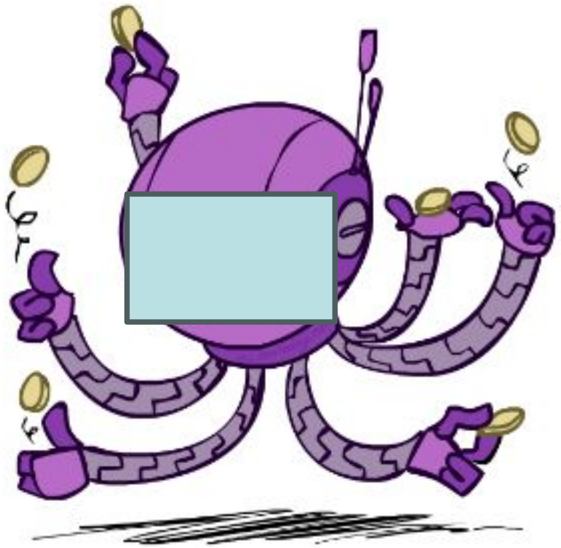
Graphical Model Notation

- Nodes: variables (with domains)
  - Can be assigned (observed) or unassigned (unobserved)
- Arcs: interactions
  - Similar to CSP constraints
  - Indicate "direct influence" between variables
  - Formally: encode conditional independence (more later)
- For now: imagine that arrows mean direct causation (in general, they don't!)



Example: Coin Flips

- N independent coin flips



X_1 $\quad$ X_2 $\quad \cdots \quad$ X_n

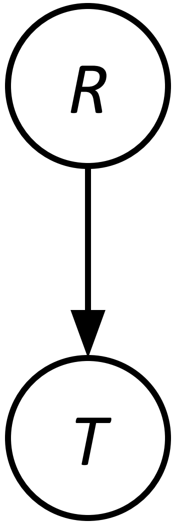
- No interactions between variables: absolute independence

Example: Traffic

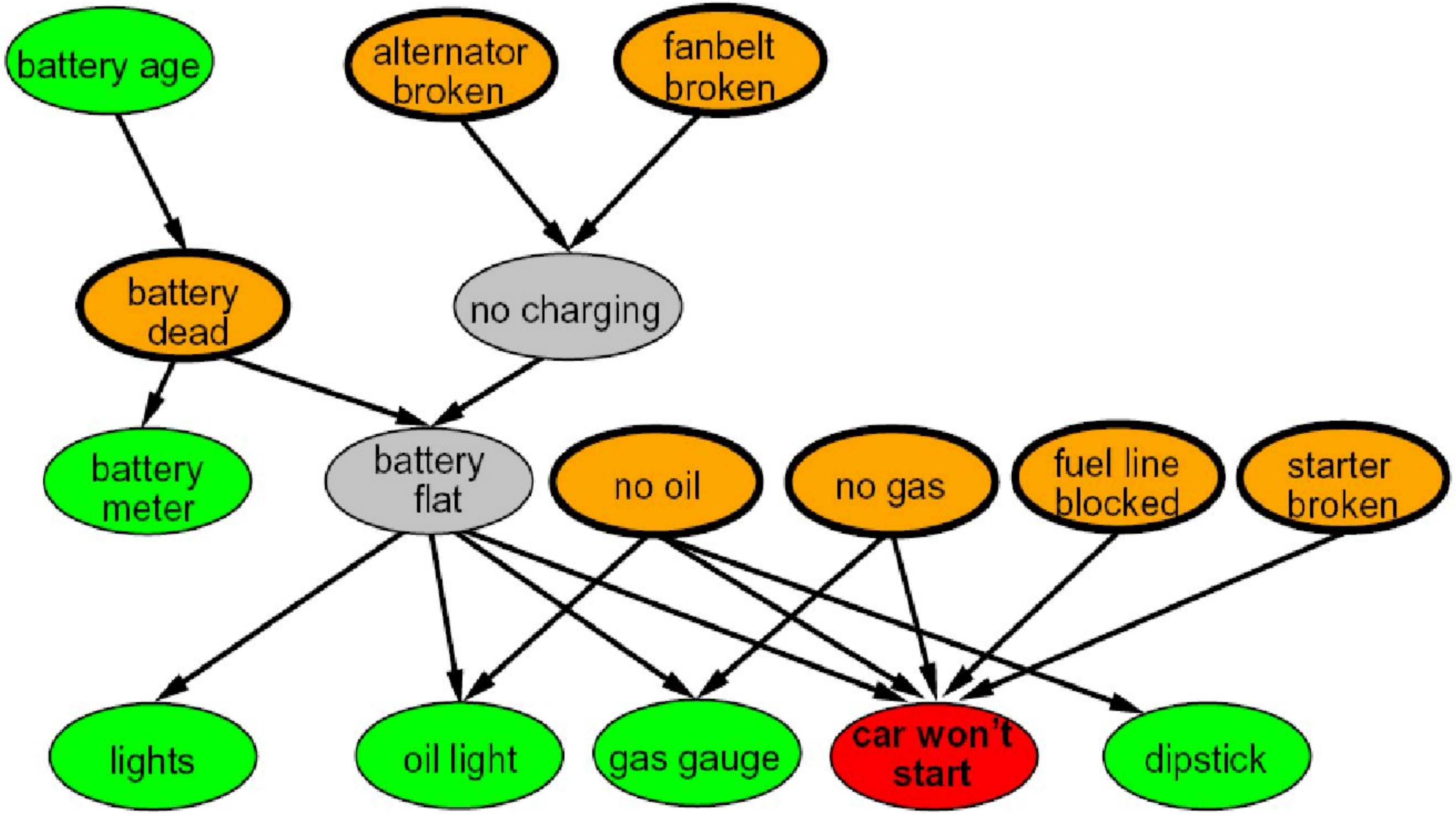
- Variables:
  - R: It rains
  - T: There is traffic

- Model 1: independence

- Model 2: rain causes traffic



Example: Car Won't Start

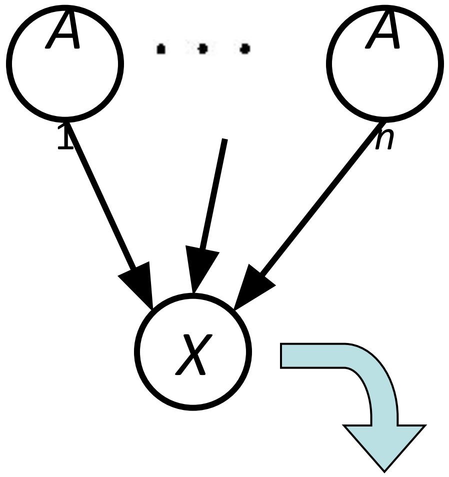


Bayes' Net Semantics

- A set of nodes, one per variable X
- A directed, acyclic graph
- A conditional distribution for each node
  - A collection of distributions over X, one for each combination of parents' values

$$P(X|a_1 \ldots a_n)$$

  - CPT: conditional probability table



$$P(X|A_1 \ldots A_n)$$

A Bayes net = Topology (graph) + Local Conditional Probabilities

Example: Coin Flips

X_1 $\quad$ X_2 $\quad \cdots \quad$ X_n

$P(X_1)$

h	0.5
t	0.5

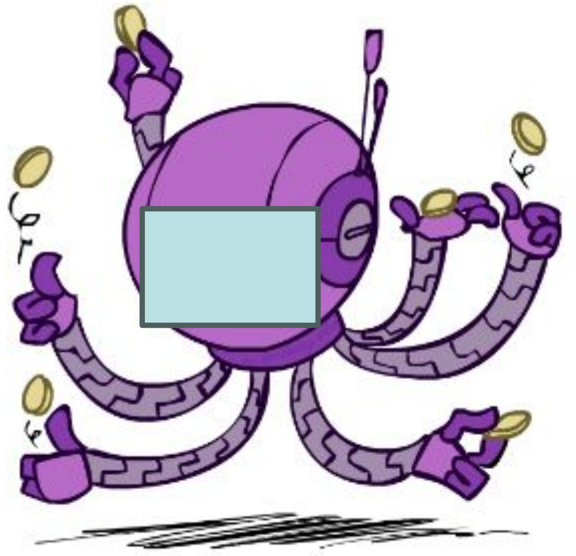
$P(X_2)$

h	0.5
t	0.5

$\cdots$

$P(X_n)$

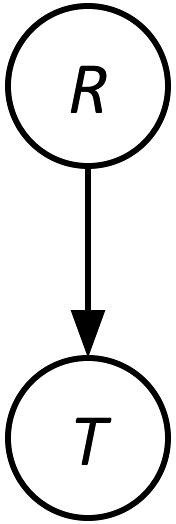
h	0.5
t	0.5



$$P(h, h, t, h) =$$

Only distributions whose variables are absolutely independent can be represented by a Bayes' net with no arcs.

Example: Traffic



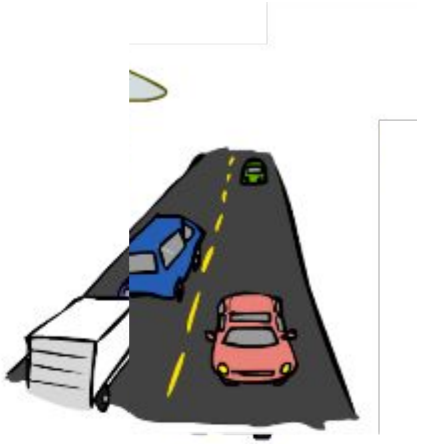
$P(R)$

+r	1/4
-r	3/4

$P(T|R)$

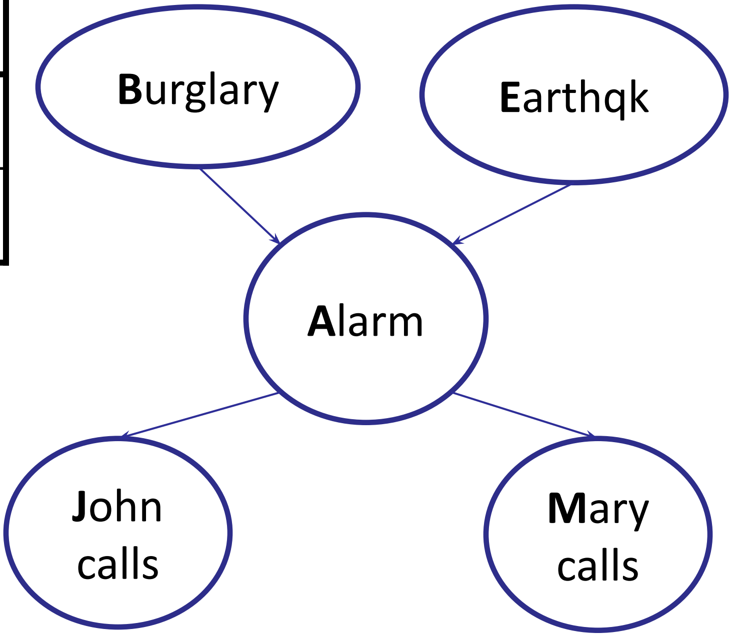
+r	+t	3/4
	-t	1/4
-r	+t	1/2
	-t	1/2

$$P(+r, -t) =$$

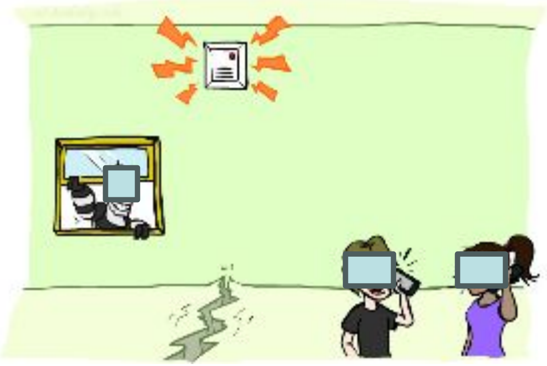


Example: Alarm Network

B	P(B)
+b	0.001
-b	0.999



E	P(E)
+e	0.002
-e	0.998



A	J	P(J|A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

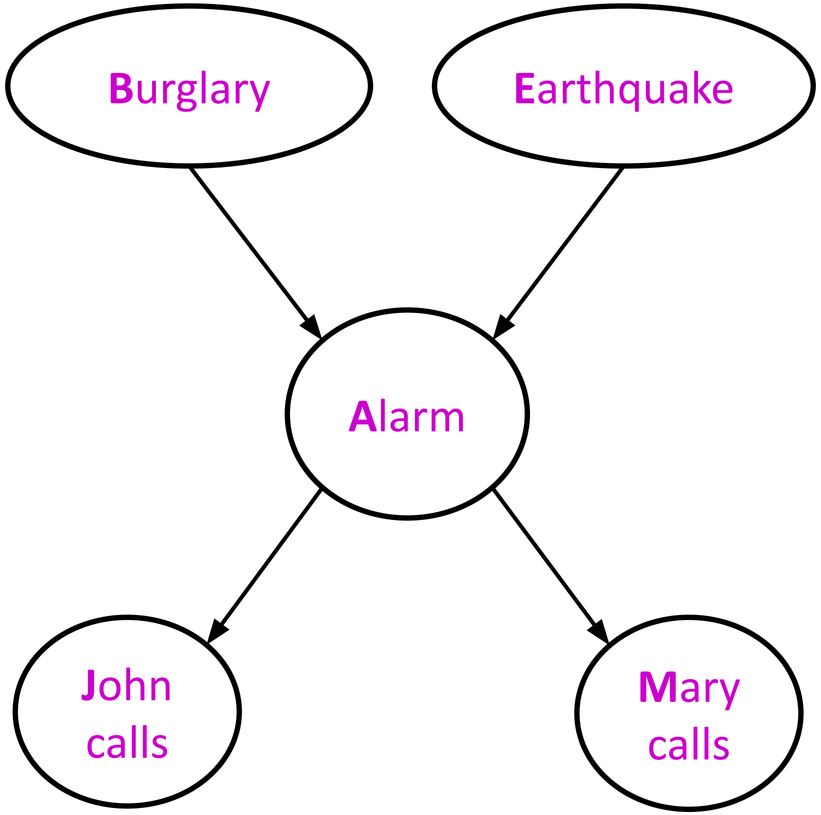
A	M	P(M|A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

B	E	A	P(A|B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

Example

P(B)	
true	false
0.001	0.999

P(E)	
true	false
0.002	0.998



B	E	P(A|B,E)	
		true	false
true	true	0.95	0.05
true	false	0.94	0.06
false	true	0.29	0.71
false	false	0.001	0.999

A	P(J|A)	
	true	false
true	0.9	0.1
false	0.05	0.95

A	P(M|A)	
	true	false
true	0.7	0.3
false	0.01	0.99

$P(b, \neg e, a, \neg j, \neg m) =$

$P(b)\ P(\neg e)\ P(a|b, \neg e)\ P(\neg j|a)\ P(\neg m|a)$

$= .001 \times .998 \times .94 \times .1 \times .3 = .000028$

Bayes net global semantics

- Bayes nets encode joint distributions as product of conditional distributions on each variable:

$$P(X_1,..,X_n) = \prod_i P(X_i \mid Parents(X_i))$$

- Exploits sparse structure: number of parents is usually small

Size of a Bayes Net

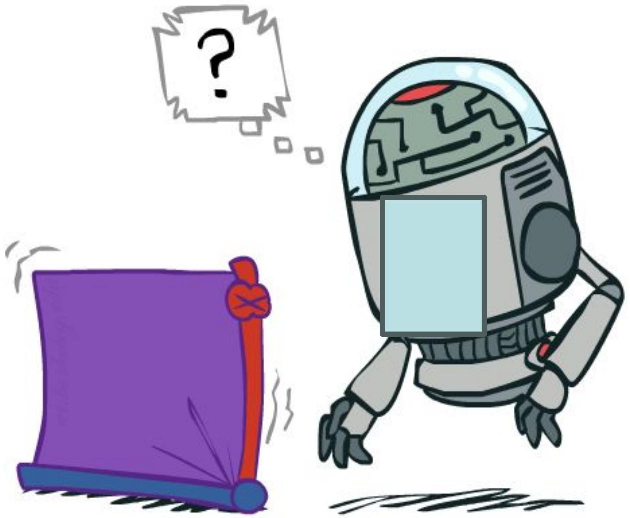
- How big is a joint distribution over N variables, each with d values?

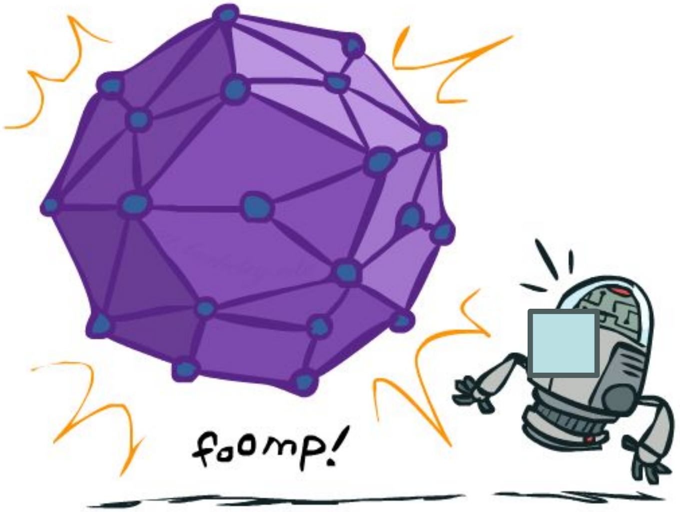
  d^N

- How big is an N-node net if nodes have at most k parents?

  $O(N * d^k)$

- Both give you the power to calculate $P(X_1, X_2, \ldots, X_N)$
- Bayes Nets: huge space savings with sparsity!
- Also easier to elicit local CPTs
- Also faster to answer queries (coming)





Conditional independence in BNs

- Compare the Bayes net global semantics

$$P(X_1,..,X_n) = \prod_i P(X_i \mid Parents(X_i))$$

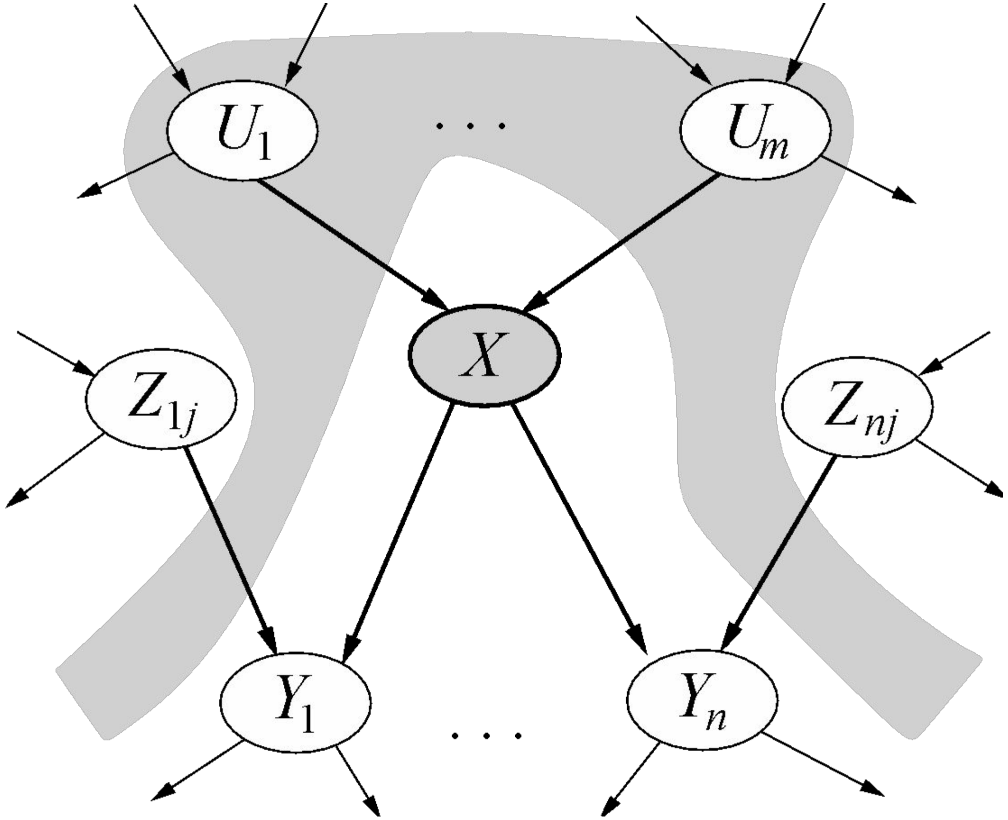
  with the chain rule identity

$$P(X_1,..,X_n) = \prod_i P(X_i \mid X_1,\ldots,X_{i-1})$$

- Assume (without loss of generality) that $X_1,..,X_n$ sorted in topological order according to the graph (i.e., parents before children), so $Parents(X_i) \subseteq X_1,\ldots,X_{i-1}$
- So the Bayes net asserts conditional independences $P(X_i \mid X_1,\ldots,X_{i-1}) = P(X_i \mid Parents(X_i))$
  - To ensure these are valid, choose parents for node X_i that "shield" it from other predecessors

Conditional independence semantics

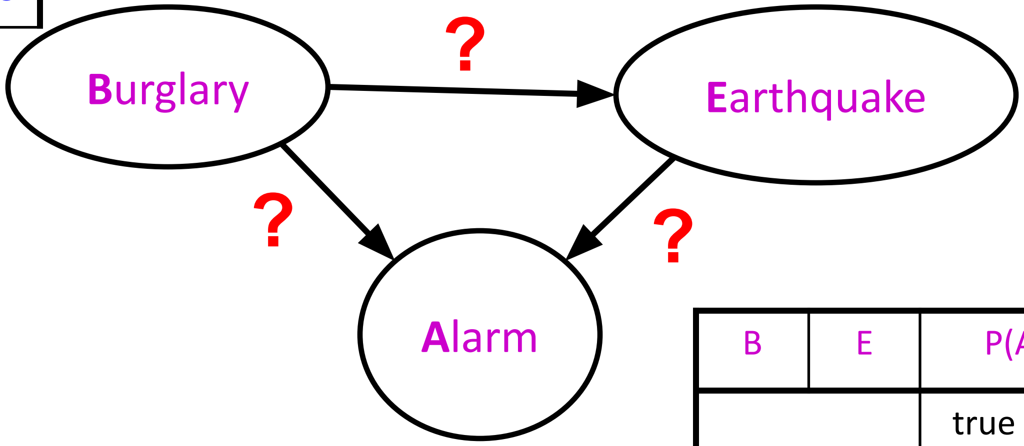
- ***Every variable is conditionally independent of its non-descendants given its parents***
- Conditional independence semantics <=> global semantics



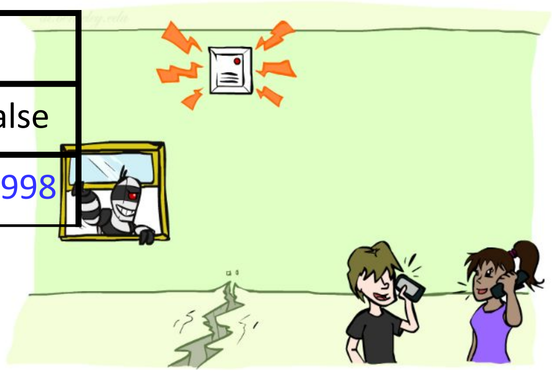
Example: Burglary

- Burglary
- Earthquake
- Alarm

P(B)	
true	false
0.001	0.999

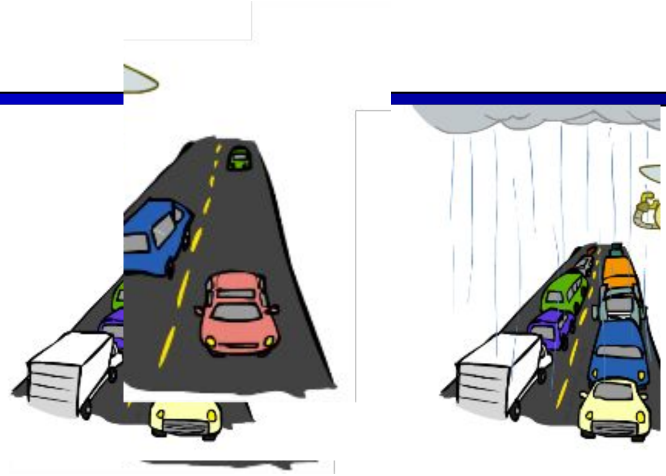


P(E)	
true	false
0.002	0.998

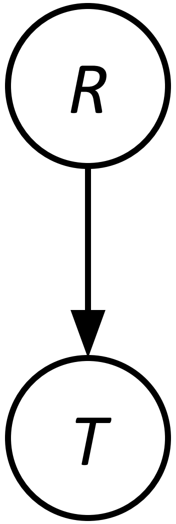


B	E	P(A|B,E)	
		true	false
true	true	0.95	0.05
true	false	0.94	0.06
false	true	0.29	0.71
false	false	0.001	0.999

Example: Traffic



direction vs Reverse Direction



$P(R)$

+r	1/4
-r	3/4

$P(T|R)$

+r	+t	3/4
	-t	1/4
-r	+t	1/2
	-t	1/2

$P(T, R)$

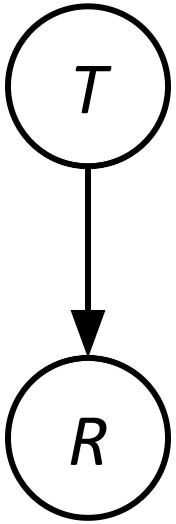
+r	+t	3/16
+r	-t	1/16
-r	+t	6/16
-r	-t	6/16

$P(T)$

+t	9/16
-t	7/16

$P(R|T)$

+t	+r	1/3
	-r	2/3
-t	+r	1/7
	-r	6/7



Causality?

- When Bayes' nets reflect the true causal patterns:
  - Often simpler (nodes have fewer parents)
  - Often easier to think about
  - Often easier to elicit from experts

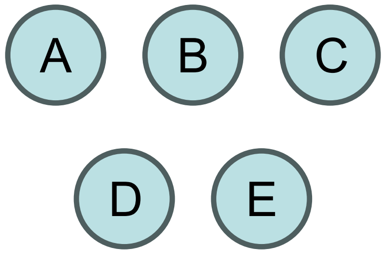
- BNs need not actually be causal
  - Sometimes no causal net exists over the domain (especially if variables are missing)
  - E.g. consider the variables *Traffic* and *Rain*
  - End up with arrows that reflect correlation, not causation

- What do the arrows really mean?
  - Topology may happen to encode causal structure
  - Topology really encodes conditional independence

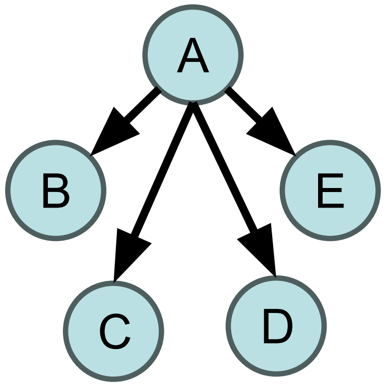
$$P(x_i|x_1, \ldots x_{i-1}) = P(x_i|parents(X_i))$$

Baye’s Net - Summary

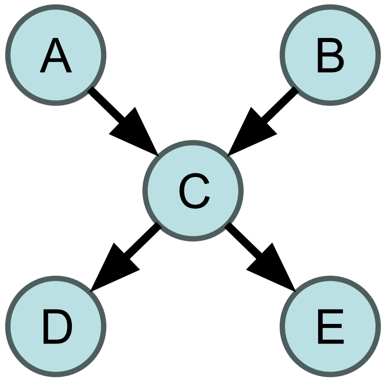
- Independence and conditional independence are important forms of probabilistic knowledge
- Bayes net encode joint distributions efficiently by taking advantage of conditional independence
  - Global joint probability = product of local conditionals
- Allows for flexible tradeoff between model accuracy and memory/compute efficiency



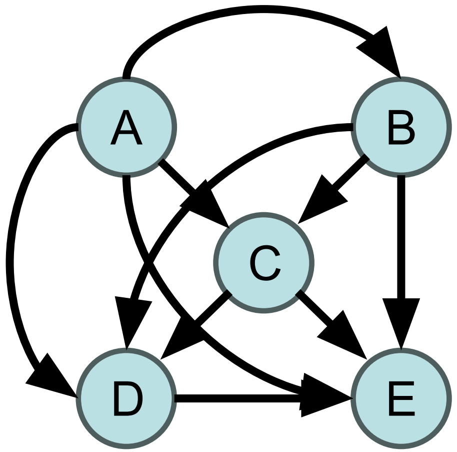
Strict Independence



Naïve Bayes



Sparse Bayes Net



Joint Distribution